ASSIGNMENT 1 – C PROGRAMMING (CO1)

Task 1: Understand the Problem and Identify the Data

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Task Marks	4 marks

1. Aim

To analyse the electricity-billing problem and identify the data the C program must store and calculate, the constants and variables it needs, the rules a correct bill must satisfy, and the assumptions used in the design.

2. Problem Analysis

Grama Jyothi Rural Electricity Co-operative needs a reliable C program to calculate household electricity bills. The program must use the previous and current meter readings to determine consumption, calculate the energy charge according to the applicable tariff, add the fixed charge, and report the total bill accurately to the paisa. The design must also consider meter rollover and the possibility of processing records for many households.

3. Inputs and why they are needed

Previous meter reading: The reading recorded at the start of the billing period. It is needed as the starting point for calculating consumption.
Current meter reading: The latest reading. Comparing it with the previous reading gives the units consumed, with special handling if the meter register has rolled over.
Household number: Identifies the consumer and keeps each bill associated with the correct household.
Household category: Identifies the applicable consumer category if the co-operative uses different tariff rules for different categories. The exact categories and their rates must come from the supplied tariff requirements.

4. Outputs and why they are needed

Units consumed: The difference between the current and previous readings, adjusted for a valid meter rollover.
Energy charge: The charge calculated from the consumed units and the applicable tariff slabs.
Fixed charge: The fixed amount charged for the billing period, according to the tariff rules.
Total bill: The amount payable, calculated as the energy charge plus the fixed charge and any other explicitly required charge.

5. Constants and variables

Values that remain unchanged during one program run—such as tariff slab boundaries, rates, the fixed charge, and the meter register's maximum value—should be represented as named constants. This makes the program easier to read and maintain. Values that vary by household or billing period—such as meter readings, units consumed, energy charge, and total bill—should be stored in variables. If tariffs change, the relevant constants can be updated in one place.

6. Values that may be negative, fractional, or very large

Valid meter readings and units consumed should not be negative. A current reading smaller than the previous reading may indicate a register rollover, but it could also indicate a reset or incorrect reading; the program should not automatically treat every decrease as a rollover without confirming the meter rules. Tariff rates or intermediate calculations may involve fractions. Monetary totals must be accurate to the paisa, so storing money as integer paise is a suitable approach. The number of households and accumulated totals can become large as processing expands from a village to a taluk or district, so the selected integer types must be checked for sufficient range.

7. Rules a correct bill must satisfy

1. Inputs must be validated before calculations are performed.
2. Consumption must be calculated correctly, including a valid meter rollover.
3. An invalid reading must not silently produce an incorrect bill.
4. The correct tariff slab and category rules must be applied.
5. The total must equal the sum of its bill components.
6. A bill must not be negative under the stated billing rules.
7. Monetary values must be calculated and displayed accurately to two decimal places.

8. Assumptions

This analysis assumes that each household has a unique identifier, meter readings are supplied in the expected register format, and the co-operative provides the applicable tariff slab limits, rates, fixed charge, category rules, and rollover behaviour. If a 16-bit unsigned meter register is specified, its values range from 0 to 65535 and it rolls over to 0 after its maximum value. A decreasing reading should be treated as rollover only when this behaviour is confirmed. No additional tariff rates or household categories are assumed beyond those supplied in the assignment.

9. Alternative concept: readings stored as text

A meter reading could be stored as a character string, such as "01530", rather than as an integer. Text can preserve leading zeros and the original displayed format, but arithmetic cannot be performed directly on the characters. The program must validate and convert the string to a number before calculating consumption. If a reading contains d digits, conversion and character-by-character validation take O(d) time. The text representation requires O(d) storage for d digits, plus string-termination storage in C. A fixed-width integer reading uses O(1) time for comparison and arithmetic and O(1) storage for a fixed-size type. Integer representation is therefore more convenient for calculations, while text can be useful for preserving the displayed reading.

10. Conclusion / Result

The problem analysis identifies the information required to produce reliable electricity bills and the rules needed to validate the results. A suitable implementation should use named tariff constants, data types with sufficient range, validated meter readings, confirmed rollover handling, and an exact approach to monetary calculations. These design choices help maintain accuracy as the number of households grows.

Evidence note: Task 1 asks for a clear written analysis. It does not provide a separate Task 1 program to execute, so a code-output screenshot is not included in this Task 1 analysis PDF. Add actual code and output evidence for tasks where the assignment requires program execution.